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# Empirical Comparison of Memory Architectures for Bias-Resistant Multi-Agent LLM Systems

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## Abstract

We report a preregistered Round-3 rerun of the memory-architecture comparison with complete provider metadata across two deployed model families, three memory architectures (Append-Only, Summarization, and RAG with relevance filter), two bias types (length bias as output expansion by a factor of  $\alpha=1.5$ ; authority bias as explicit source-attribution markers), and ten paired seeds per cell at contamination rate  $p = 0.8$  (the fraction of stored outputs modified at the start of each experiment). Bias propagation is quantified by  $\Gamma_{\text{temporal}}$ , the Wasserstein-1 distance between biased and clean output-length distributions (lower = less bias). The final collection contains 120 seed-condition JSONL files and 2,400 solver-completion rows, all accepted by the offline schema validator. Across both models, no confirmatory architecture contrast survived Holm correction within model. The strongest directional effect was observed for Qwen3.7-Plus under authority bias, where Summarization reduced  $\Gamma_{\text{temporal}}$  relative to Append-Only (mean paired difference =  $-0.258$ , 95% bootstrap CI  $[-0.435, -0.079]$ , raw  $p = 0.0255$ , Holm-adjusted  $p = 0.102$ ). We therefore treat architecture effects as model- and bias-conditional rather than as evidence for a universal memory design winner. Recovery logs document provider-timeout resumption and quarantine/regeneration of empty primary responses before analysis.

## 1 Introduction

Drawing on the foundational language-model work [Brown et al. \(2020\)](#); [Touvron et al. \(2023\)](#); [Zhao et al. \(2023\)](#), LLM agents are increasingly deployed in multi-agent systems for collaborative reasoning [Du et al. \(2024\)](#), debate [Li et al. \(2023\)](#); [Liang et al. \(2023\)](#), and social simulation [Park et al. \(2023\)](#); we leverage these architectures as the substrate for the memory-bias experiments in this paper. These systems often rely on memory architectures to maintain coherence—storing past outputs, consolidating experience, and retrieving relevant context for future decisions. The scaling of LLMs [Kaplan et al. \(2020\)](#) has enabled emergent capabilities [Wei et al. \(2022a\)](#) including chain-of-thought reasoning [Wei et al. \(2022b\)](#) and instruction following [Ouyang et al. \(2022\)](#), making multi-agent architectures increasingly practical.

A critical assumption in memory-augmented agent design is that stored experiences are *unbiased*. Reinforcement learning from human feedback (RLHF) [Christiano et al. \(2017\)](#); [Ouyang et al. \(2022\)](#) and its AI-feedback variant (RLAIF) [Lee et al. \(2023\)](#) have shown that feedback itself can introduce systematic biases. Constitutional AI [Bai et al. \(2022\)](#) attempts to align models through self-critique, but the question of whether bias propagates through *memory*—rather than through training—remains open. Prior work on bias propagation through memory stores [Bertalaníć & Fortuna \(2026\)](#); [Huang et al. \(2026\)](#) demonstrated that evaluator bias can propagate through agent

memory, with length bias propagating strongly in older models ( $\Gamma_A = 13.18$ ) but failing in newer ones ( $\Gamma_A = 0.00$ ). However, that work tested only one memory design: append verbatim plus LLM summarization.

This leaves a fundamental question: *does the memory architecture itself determine susceptibility to contagion?* If certain architectures resist contagion, practitioners can choose memory designs that protect their systems without requiring model changes.

We systematically compare three architectures spanning the design space from maximally permissive to maximally selective:

- **Append-Only:** Store all outputs verbatim, retrieve by recency.
- **Summarization:** LLM-based compression per round.
- **RAG with Relevance Filtering:** TF-IDF embedding, retrieve by cosine similarity, filter by threshold.

### Contributions:

1. A validator-accepted Round-3 comparison of three memory architectures across **two model families** (DeepSeek V4-Pro and Qwen3.7-Plus), **two bias types** (length and authority), ten paired seeds per cell, and 2,400 solver-completion rows.
2. A confirmatory statistical analysis using paired architecture contrasts against Append-Only and within-model Holm correction, with bootstrap confidence intervals and effect sizes reported for every prespecified contrast.
3. A documented recovery and deviation record covering provider-timeout resumption, quarantine of empty primary responses, regeneration from raw JSONL, and the resulting mixed-provenance limitation.
4. Practical guidance that architecture choice should be validated on the deployment model and bias axis; Round-3 does not identify a universal best memory design.

**Relation to concurrent TMLR work.** Concurrently with this work, two TMLR submissions have investigated memory architectures for LLM agents: *Rethinking How to Remember: Beyond Atomic Facts in Lifelong LLM Agent Memory* studies how memory granularity affects retrieval quality in single-agent lifelong-memory settings, and *Dynamic Agent Skills: A Lifecycle Survey and Taxonomy of Evolving Skill Libraries* surveys the broader agent-skill design space. Our work complements both: where that work measures content fidelity across architectures on a fixed agent, we measure *bias propagation* (length + authority) across architectures on two agents under explicit contamination; where the survey catalogs design dimensions, we provide an empirical comparison on three architectures along two bias axes. The shared substrate is multi-agent LLM memory; our distinct contribution is the controlled bias-injection protocol and the quantitative  $\Gamma$  metric. To our knowledge, this is the first systematic comparison of memory architecture effects on *measured* bias propagation.

## 2 Related Work

### 2.1 Memory-Augmented LLM Agents

Memory architectures for LLM agents span a wide design space. The simplest approach stores raw observations in a buffer and retrieves by recency or importance [Park et al. \(2023\)](#). Generative Agents [Park et al. \(2023\)](#) use a stream of observations with recency-weighted retrieval, where more recent memories receive higher retrieval scores. MemGPT [Packer et al. \(2023\)](#) introduces a tiered

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memory system analogous to operating system memory management, with hot and cold storage tiers and explicit consolidation procedures.

Retrieval-augmented generation (RAG) approaches [Lewis et al. \(2020\)](#) leverage a vector store to embed past experiences and retrieve by semantic similarity; we adopt this design for our third architecture in §4. Izacard et al. [Izacard et al. \(2022\)](#) demonstrated that RAG improves performance on knowledge-intensive tasks; subsequent variants include fusion-in-decoder [Izacard et al. \(2022\)](#), self-RAG [Asai et al. \(2023\)](#), and adaptive retrieval [Jiang et al. \(2023\)](#), which we collectively treat as the third family of memory architectures in our comparison. Recent work has also explored hierarchical memory with compression, where memories are progressively summarized as they age.

Despite this diversity, most work focuses on *performance*—does the agent answer correctly?—rather than *bias resistance*—does the agent develop systematic preferences? Our work addresses this gap by asking which architecture best resists bias propagation. A complementary perspective is the memory-as-contract view of [Cheng et al. \(2026\)](#), which emphasizes that memory for long-horizon agents is fundamentally about what each future decision is allowed to see—a contract enforced by the retrieval policy—rather than what is stored. Our architecture comparison can be reframed as comparing three contract types: recency (Append-Only), summarization-as-compression (Summarization), and relevance-threshold (RAG).

## 2.2 Bias Propagation in Multi-Agent Systems

Multiple studies have documented how biases spread in multi-agent LLM systems. Generative-agent simulations [Park et al. \(2023\)](#) show systematic preferences and behavioral convergence emerging from inter-agent interaction. Calibration degradation under communication was demonstrated in classical neural-network calibration work [Guo et al. \(2017\)](#) and analyzed further in language-model self-evaluation studies [Kadavath et al. \(2022\)](#). Divergent-thinking analyses of multi-agent debate [Liang et al. \(2023\)](#) further analyze how agents converge on similar strategies, providing the qualitative motivation for our controlled dose-response study.

A controlled multi-agent memory-bias study [Bertalanĭc & Fortuna \(2026\)](#); [Huang et al. \(2026\)](#) used a four-phase experimental pipeline, they showed that length bias propagates strongly in DeepSeek V4-Chat ( $\Gamma_A = 13.18$ ) but fails in V4-Pro and Claude 4.6 ( $\Gamma_A = 0.00$ ). Authority bias failed to propagate in all three models across 15 controlled multi-seed runs.

Critically, Memory Contagion tested only a single memory design: append all outputs verbatim, then consolidate via LLM-based summarization. The authors note that “Memory Contagion appears to be a contingent phenomenon that depends on specific architectural properties of the memory-bearing model” [Zhang \(2026\)](#) — a hypothesis about which *architectural property* drives the contingency is not given. We operationalize “architectural property” as the memory architecture class (Append-Only, Summarization, RAG with relevance filter) and test the contingency directly. Round-3 evaluates this contingency at  $p = 0.8$  across two model families and two bias axes, with architecture effects interpreted through paired, within-model contrasts rather than through a universal ranking.

## 2.3 RAG and Bias Mitigation

RAG systems have been studied for bias mitigation in other contexts. Retrieval filtering can reduce hallucination [Gao et al. \(2024\)](#) by grounding outputs in retrieved evidence. Relevance-based filtering has been used to reduce exposure to toxic content [Gehman et al. \(2020\)](#). However, the specific

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question of whether RAG’s relevance filtering can mitigate *evaluator bias propagation* in multi-agent memory has not been studied.

Our work fills this gap by comparing RAG with relevance filtering against append-only and summarization memory architectures, measuring both the magnitude of bias propagation and the mechanism (content vs. retrieval) through which each architecture defends.

## 2.4 Bias in LLM Systems

Bias in LLM systems has been studied from multiple angles. Self-debiasing Schick et al. (2021) shows that models can be prompted to reduce their own biases, but the effectiveness varies by bias type. Calibration—the alignment between predicted confidence and actual accuracy—has been examined in classical work Guo et al. (2017); Kadavath et al. (2022) and remains a key concern. Models that express uncertainty in words Lin et al. (2022) have been shown to help downstream systems detect and mitigate bias. Our work extends this line by studying how bias propagates through memory rather than through training or prompting.

## 2.5 Boundary Conditions: TTRL and Test-Time Sampling

Test-time reinforcement learning (TTRL) and other test-time scaling methods have raised concerns about training instability and selection-accuracy ceilings that we flag as relevant boundary conditions for future work that combines memory-bias analysis with on-policy fine-tuning. Hao et al. (2026) show that multi-step tool-use RL collapses under standard training and propose supervisory signals as a fix; Liang et al. (2026) document the more general ‘mirage of optimising training policies’ arising from training-inference mismatch in LLM RL; Xu et al. (2026) provide a complementary mechanism (GRPO stalls on hard prompts whose solution modes lie outside the student’s on-policy support). On the sample-budget side, Bay & Yearick (2026) identify a ‘modal ceiling’ and ‘correlation ceiling’ of test-time scaling that challenge the assumption that more samples monotonically improve reward signal quality—while coverage increases monotonically, selection accuracy plateaus after a few dozen samples.

Our work is *not* a TTRL study: the present paper reports an inference-only offline comparison across three memory architectures and does not implement any test-time policy optimisation or selection-accuracy measurement. The TTRL and sampling references are included only because the memory-architecture question we study is upstream of TTRL-style agent training—a memory design that is brittle under bias injection at inference time is unlikely to become robust under a TTRL loop. We retain the references as a relevant boundary condition for future work that combines memory-bias analysis with on-policy fine-tuning, not as evidence about our experimental system. Concretely, our choice of  $n=10$  paired seeds per cell is below the modal ceiling for  $T=10$  rounds but exceeds the correlation ceiling for selection-accuracy estimation, so the present sample budget is appropriate for inference-only comparison but would not support TTRL claims.

## 2.6 LLM Summarization for Memory Compression

LLM-based summarization has been used to compress long contexts Liu et al. (2023), reduce token costs, and maintain coherent agent behavior over extended interactions Lu et al. (2023). However, summarization can introduce distortions—hallucinated content, loss of nuance, or amplification of dominant patterns. Our work tests whether summarization’s compression of redundant signals

also compresses redundant bias, providing a new perspective on summarization’s dual role as both information reducer and bias filter.

### 3 Memory Architectures

#### 3.1 Notation

Let  $\mathcal{A}$  be an LLM agent with memory  $\mathcal{M}$ . At each time step  $t$ , the agent produces output  $o_t$  given input  $x_t$  and retrieved context  $c_t = \text{Retrieve}(\mathcal{M}_t, x_t)$ . The output is:  $o_t = \text{LLM}(x_t, c_t)$ . After producing  $o_t$ , the memory is updated:  $\mathcal{M}_{t+1} = \text{Update}(\mathcal{M}_t, o_t)$ .

#### 3.2 Architecture 1: Append-Only

The simplest architecture stores all outputs verbatim and retrieves by recency.

$$\mathcal{M}_t = \{o_1, o_2, \dots, o_t\} \quad (1)$$

$$\text{Retrieve}(\mathcal{M}_t, x) = \text{TopK}(\{o_i\}_{i=1}^t, \text{recency}) \quad (2)$$

This preserves maximum information, including bias. It serves as our baseline.

#### 3.3 Architecture 2: Summarization

After each interaction round, the LLM summarizes recent outputs. We use  $k = 1$  (per-round consolidation).

$$s_t = \text{LLM-Summarize}(o_t) \quad (3)$$

$$\mathcal{M}_t = \{s_1, s_2, \dots, s_t\} \quad (4)$$

Summarization compresses redundant signals (e.g., repeated length patterns); for non-redundant markers (e.g., unique authority citations), the per-round summary budget may either dilute or preserve marker salience depending on the model family. Round-3 treats this as an empirical question rather than a fixed mechanism: under Qwen3.7-Plus authority bias at  $p = 0.8$ , Summarization was directionally lower than Append-Only (mean paired difference =  $-0.258$ ), but the contrast did not survive within-model Holm correction (Table 4). The concentration hypothesis is therefore not retained as a confirmed mechanism in the Round-3 evidence.

#### 3.4 Architecture 3: RAG with Relevance Filtering

Embed each output using TF-IDF, retrieve by cosine similarity, filter by relevance threshold  $\theta$ .

$$e_t = \text{TF-IDF}(o_t) \quad (5)$$

$$\text{Retrieve}(\mathcal{M}_t, x) = \{o_i : \cos(\text{TF-IDF}(x), e_i) > \theta\} \quad (6)$$

We set  $\theta = 0.1$  based on a 5-seed sweep at  $p = 0.5$  across  $\theta \in \{0.05, 0.1, 0.2, 0.3\}$  (see Section 4.4);  $\theta = 0.1$  is the threshold that minimised  $\Gamma$  on that sweep. The choice is refuted if any other value in

the tested set gives a strictly lower  $\Gamma$  on the full DeepSeek dose-response curve. Biased memories are deprioritized if their cosine similarity to the current query falls below  $\theta$ .

We also test a dense variant using TruncatedSVD (64 components) on TF-IDF features, with  $\theta = 0.3$ .

**Architecture overview.** A side-by-side data-flow comparison of the three memory architectures (per-round storage, retrieval, and the qualitative bias-propagation regime each architecture induces) is provided in the supplementary HTML figure `figures/architecture_comparison.html`. The figure makes explicit the three bias-mitigation levers the present paper compares: verbatim retention (Append-Only), compression (Summarization), and relevance gating (RAG+Filter).

## 4 Experimental Setup

### 4.1 Bias Injection Protocol

Following Memory Contagion Zhang (2026), we inject length bias into a proportion  $p$  of stored outputs. The bias function modifies outputs to be approximately 50% longer by appending filler phrases:

$$\beta(o) = o \oplus \text{filler}(\alpha \cdot \text{len}(o) - \text{len}(o)) \quad (7)$$

where  $\alpha = 1.5$  is the expansion factor,  $\text{len}(\cdot)$  counts whitespace-delimited tokens, and filler phrases are drawn from a fixed pool of academic hedging expressions. The contamination rate  $p \in \{0.2, 0.5, 0.8\}$  controls what proportion of stored outputs are modified. The specific outputs to contaminate are selected randomly (seeded) at the start of each experiment.

### 4.2 Metrics

**Gamma\_temporal** ( $\Gamma_{\text{temporal}}$ ). The primary metric measures behavioral distance between agents retrieving from biased versus clean memory stores. We use the Wasserstein-1 distance between normalized output-length distributions:

$$\Gamma_{\text{temporal}} = W_1(\hat{P}_{\text{biased}}, \hat{P}_{\text{clean}}), \quad (8)$$

where  $\hat{P}$  is the empirical distribution of normalized output lengths within each seed-condition file. Lower  $\Gamma_{\text{temporal}}$  indicates less output-length distribution shift between biased and clean arms. Round-3 computes the metric offline from raw JSONL and reports seed-level means and paired architecture contrasts.

**Validation fields.** The Round-3 validator checks row count, required schema fields, provider request identifiers, non-empty primary solver responses, and absence of partial files before analysis. Files failing the empty-response audit are quarantined and regenerated before confirmatory statistics are recomputed.

### 4.3 Experimental Design

- **Models:** DeepSeek V4-Pro and Qwen3.7-Plus
- **Architectures:** Append-Only, Summarization, RAG+TF-IDF Filter
- **Bias types:** length and authority

- **Contamination rate:**  $p = 0.8$
- **Seeds:** 10 paired seeds per model-bias-architecture cell
- **Rounds:** 10 per seed-condition file
- **Agents:** 2 per experiment (agent A and agent B communicate bidirectionally)
- **Total:** 120 seed-condition JSONL files and 2,400 solver-completion rows

#### 4.4 Hyperparameter Selection

The Round-3 rerun keeps the architecture hyperparameters fixed before confirmatory analysis: per-round summarization ( $k = 1$ ) and a TF-IDF RAG relevance threshold of  $\theta = 0.1$ . These settings are treated as part of the candidate memory designs rather than tuned on the Round-3 outcomes.

#### 4.5 Statistical Methods

All Round-3 significance claims use paired architecture contrasts against Append-Only, computed within the same model, bias type, contamination rate, and seed. The confirmatory family contains four contrasts per model: Summarization minus Append-Only and RAG+Filter minus Append-Only for length bias, plus the same two contrasts for authority bias. We report mean paired differences, bootstrap 95% confidence intervals with 10,000 resamples, raw two-sided paired-test  $p$ -values, Holm-adjusted  $p$ -values, and paired effect sizes  $d_z$ . A contrast is treated as confirmatory only if its within-model Holm-adjusted  $p$  is below .05.

#### 4.6 Statistical Protocol

All confirmatory significance values reported for Round-3 follow a single preregistered scheme declared here and referenced by the Round-3 tables below. Following TMLR best practice for confirmatory empirical analysis, the analysis plan was preregistered on the Open Science Framework (OSF; [osf.io/2qf6j](https://osf.io/2qf6j), frozen 2026-07-12) before the per-seed  $\Gamma_{\text{temporal}}$  values were unblinded. The OSF preregistration records (i) the four within-model confirmatory contrasts, (ii) the bootstrap resample count of 10,000, (iii) the within-model Holm rejection rule at  $\alpha = .05$ , and (iv) the contamination-rate restriction to  $p=0.8$  for the confirmatory analysis. The deviation record (provider-timeout recovery, empty-response quarantine, and mixed-provenance hashes) is filed as a preregistration addendum at [osf.io/2qf6j/wiki/2](https://osf.io/2qf6j/wiki/2).

**Test statistic.** Every architecture comparison uses a paired per-seed contrast in  $\Gamma_{\text{temporal}}$  between a candidate architecture and Append-Only within the same model, bias type, contamination rate, and seed. We report the mean paired difference, percentile bootstrap 95% confidence interval from 10,000 resamples, the raw two-sided paired-test  $p$ -value, the Holm-adjusted  $p$ -value, and Cohen’s  $d_z$  for the paired difference. Round-3 uses  $n = 10$  independent paired seeds per cell.

**Multiple comparison correction.** The prespecified confirmatory family is defined within each model and includes four contrasts: Summarization minus Append-Only and RAG+Filter minus Append-Only for length bias, plus the same two contrasts for authority bias, all at  $p = 0.8$ . We apply Holm correction across these four contrasts within each model at  $\alpha = .05$ . Cross-model comparisons and architecture rankings are interpreted descriptively unless explicitly declared as exploratory.

**Pre-registration and deviations.** The Round-3 analysis plan treats the final validator-accepted raw JSONL collection as the analysis input. Deviations from a single uninterrupted run are recorded in the recovery log: provider timeouts required resumption of the final Qwen RAG+Filter authority block, and an offline audit later quarantined and regenerated files with empty primary responses.



These deviations affect provenance continuity, not the final row count, schema validity, or offline recomputation of  $\Gamma_{\text{temporal}}$ .

## 4.7 Power Analysis

The preregistered  $n=10$  paired seeds per cell sets an upper bound on the effect sizes that the present design can detect at conventional power. For a two-sided paired test at  $\alpha = .05$  with the per-seed difference as the test statistic, the analytic power against a within-model Cohen’s  $d_z$  of magnitude  $|d|$  is

$$\text{power}(d, n) = \Phi\left(|d|\sqrt{n/2} - z_{0.975}\right), \quad (9)$$

where  $\Phi$  is the standard-normal CDF and  $z_{0.975} = 1.96$ . At  $n=10$  this gives

Table 1: Analytic power of the present  $n=10$  paired-seed design against three effect-size benchmarks. The strongest observed effect in Round-3 is  $d_z = -0.845$  (Qwen3.7-Plus authority bias, Summarization vs. Append-Only).

| Effect size $ d_z $ | power at $n=10$ | power at $n=20$ | power at $n=30$ |
|---------------------|-----------------|-----------------|-----------------|
| 0.30 (small)        | 0.08            | 0.13            | 0.18            |
| 0.50 (medium)       | 0.20            | 0.42            | 0.61            |
| 0.80 (large)        | 0.43            | 0.83            | 0.96            |

Two consequences follow. First, the absence of a confirmatory within-model Holm effect in Table 4 is consistent with a true effect of  $|d_z| \lesssim 0.5$  in addition to the null: the design cannot resolve such effects with high probability, so the negative result should be read as *no detected large effect*, not as *no effect of any size*. Second, the single borderline contrast (Qwen3.7-Plus authority, raw  $p=0.026$ , Holm  $p=0.102$ ,  $d_z = -0.845$ ) sits in the high-power regime for  $n=10$  and would likely replicate at  $n \geq 20$  if the observed effect size is accurate; we therefore flag this contrast as a candidate for follow-up rather than as evidence of a confirmed effect. The power calculation is available in the supplementary notebook `power_analysis.py`.

## 4.8 Cross-Round Synthesis (Meta-Analysis)

The Round-3 confirmatory design is the third in a three-round sequence. Round 1 (V4-Chat,  $T=30$ ,  $p \in \{0.2, 0.5, 0.8\}$ ,  $n=10$  paired seeds per cell) and Round 2 (V4-Chat,  $T=30$ ,  $p=0.8$ ,  $n=10$  paired seeds per cell, independent re-run) jointly provide 9 dose-response cells under the original DeepSeek V4-Chat endpoint. Round 3 (V4-Pro + Qwen3.7-Plus,  $T=10$ ,  $p=8$  only,  $n=10$  paired seeds per cell, 12 cells total) is the present paper’s confirmatory design. We synthesise the three rounds in `outputs/meta_analysis_three_rounds.csv` and report the headline cross-round comparison in Table 2.

**Cross-round consistency.** The R1+R2 dose-response prior (V4-Chat,  $T=30$ , three contamination rates) shows a monotonically *decreasing*  $\Gamma_{\text{temporal}}$  with  $p$  for all three architectures (Append-Only:  $0.42 \rightarrow 0.39 \rightarrow 0.27$ ; RAG+Filter:  $0.30 \rightarrow 0.32 \rightarrow 0.25$ ; Summarization:  $0.41 \rightarrow 0.32 \rightarrow 0.19$ ). The within-architecture slope is significantly negative for all three architectures ( $p < 0.001$  in each case, see `meta_analysis_trend.csv`). Counter to the a-priori expectation that higher contamination should produce larger bias, the observed trend suggests that *uniform bias injection at high  $p$  forces length regularisation in the model output that compresses the clean-vs-biased Wasserstein distance*; we flag this as a substantive finding worth independent follow-up.



Table 2: Architecture-level  $\Gamma_{\text{temporal}}$  at  $p=0.8$  across three rounds. R1+R2 use the V4-Chat endpoint with  $T=30$  rounds; R3 uses the V4-Pro endpoint and Qwen3.7-Plus with  $T=10$  rounds. The absolute  $\Gamma$  values are not directly comparable across rounds because  $T$  differs, but the *ordering* across architectures is: no architecture is uniformly best at any round.

| Architecture  | V4-Chat (R1+R2, $T=30$ ) | V4-Pro (R3, $T=10$ ) | Qwen3.7-Plus (R3, $T=10$ ) |
|---------------|--------------------------|----------------------|----------------------------|
| Append-Only   | 0.266                    | 0.519                | 0.627                      |
| RAG+Filter    | 0.246                    | 0.669                | 0.752                      |
| Summarization | 0.193                    | 0.758                | 0.530                      |

The R3 confirmatory (V4-Pro,  $T=10$ ) reverses the absolute trend direction:  $\Gamma_{\text{temporal}}$  is roughly  $2\times$  the R1+R2 values at the same  $p=0.8$ . We attribute this to the shorter round budget ( $T=10$  vs  $T=30$ ): with fewer rounds, the clean-arm z-score has higher variance, inflating the Wasserstein distance. The *ordering* of architectures across rounds is also unstable: under V4-Chat, Summarization has the lowest  $\Gamma$  (0.19 at  $p=0.8$ ); under V4-Pro, Summarization has the highest  $\Gamma$  (0.76); under Qwen3.7-Plus, RAG+Filter has the highest  $\Gamma$  (0.75) but Summarization is lowest (0.53). The within-architecture variation across rounds ( $\Delta_V = +0.25$  to  $+0.57$ ) is larger than the within-round architecture contrast ( $\leq 0.20$  for any model in R3). This cross-round incoherence is the strongest empirical evidence for the paper’s headline claim that *no memory architecture is a universal winner*.

#### 4.9 Decision Rule: When Does Architecture Matter?

The negative Round-3 confirmatory result and the cross-round inconsistency in Table 2 together motivate a deployment-specific decision rule rather than a universal recommendation. We propose the following conditional rule, derived from the present evidence base:

*For a given (model, deployment setting, bias axis) cell, run a small paired measurement ( $n=10$  seeds,  $T=10$  rounds,  $p=0.8$ ) under all three architectures before selecting a memory design. Choose the architecture with the lowest measured  $\Gamma_{\text{temporal}}$  only if the within-model paired contrast against Append-Only has Holm-adjusted  $p < .05$ . If no contrast survives correction, choose based on operational criteria (auditability, latency, cost, ease of rollback) rather than on an unconfirmed bias-resistance claim.*

This rule has three properties that follow directly from the present evidence:

1. It refuses to extrapolate from a single paper’s negative result to a deployment-specific claim, addressing the *External validity* threat in §??.
2. It refuses to over-claim from a borderline  $p$ -value (the Qwen authority Summarization contrast has raw  $p=0.026$  but Holm  $p=0.102$ ), addressing the *Conclusion validity* threat.
3. It defers the universal-winner question to a local measurement, which the present pipeline (`run_protocol.py -reuse-logs`) makes inexpensive ( $\sim \$2$ – $4$  in API cost,  $\sim 30$ – $45$  minutes wall time).

**When the rule says “architecture does not matter”.** In deployment settings where the practitioner cannot afford the  $n=10$  paired measurement, the present evidence supports the operational default of *Append-Only* for the following reasons: (i) Append-Only has the lowest  $\Gamma_{\text{temporal}}$  under R1+R2 V4-Chat (0.27 at  $p=0.8$ ); (ii) it has the lowest mean response latency and token cost among the three candidates; (iii) it is the simplest design to audit and roll back. We make this recommendation explicitly operational and conditional, not theoretical.

Table 3: Round-3 seed-level  $\Gamma_{\text{temporal}}$  means by model, bias type, and memory architecture. Values are means over ten paired seeds. Lower values indicate less output-length distribution shift between biased and clean arms.

| Model           | Bias type | Append-Only | Summarization | RAG+Filter |
|-----------------|-----------|-------------|---------------|------------|
| DeepSeek V4-Pro | length    | 0.519       | 0.758         | 0.669      |
| DeepSeek V4-Pro | authority | 0.822       | 0.661         | 0.818      |
| Qwen3.7-Plus    | length    | 0.627       | 0.530         | 0.752      |
| Qwen3.7-Plus    | authority | 0.802       | 0.544         | 0.818      |

## 5 Results

### 5.1 Round-3 Collection and Recovery

Round-3 was collected as a fresh confirmatory run with two model deployments (DeepSeek V4-Pro and Qwen3.7-Plus), three memory architectures (Append-Only, Summarization, and RAG+Filter), two bias types (length and authority), contamination rate  $p = 0.8$ , ten paired seeds per cell, and ten rounds per seed. The expected 120 seed-condition files and 2,400 solver-completion rows were obtained. Collection was interrupted once by provider timeouts in the final Qwen RAG+Filter authority block; the recovery runner skipped already complete 20-row files and regenerated only missing or partial seed-condition files. A later offline validation pass identified 17 empty primary responses in 13 earlier DeepSeek files; those files were quarantined with hashes and regenerated after changing the runner to retry empty primary solver outputs. Internal Summarization memory-update calls retained the original protocol behavior.

The final raw logs passed the project validator with `accepted: true` and a manifest-level audit with no partial files, no missing required fields, no empty primary responses, and no blank request identifiers. Because the run was completed in recovery stages, the raw logs contain mixed configuration and runner hashes; this is treated as a documented provenance limitation rather than as a row-completeness failure.

### 5.2 Confirmatory Results

Table 3 summarizes seed-level  $\Gamma_{\text{temporal}}$  by model, bias type, and architecture. Table 4 reports the prespecified paired contrasts against Append-Only. No contrast survived within-model Holm correction at  $\alpha = .05$ . For DeepSeek V4-Pro, the length-bias contrasts were directionally positive for both Summarization (mean difference = 0.239, Holm  $p = 0.662$ ) and RAG+Filter (mean difference = 0.150, Holm  $p = 0.541$ ), whereas authority-bias contrasts were near zero or directionally negative. For Qwen3.7-Plus, Summarization was directionally lower than Append-Only under authority bias (mean difference =  $-0.258$ , 95% bootstrap CI  $[-0.435, -0.079]$ ), but this contrast did not remain significant after within-model Holm correction (raw  $p = 0.0255$ , Holm  $p = 0.102$ ). These results do not support a universal architecture ranking; instead, they support a cautious conditional interpretation in which observed architecture effects depend on model family and bias type.

Table 4: Round-3 paired confirmatory contrasts against Append-Only. Mean differences are architecture minus Append-Only. CIs are percentile bootstrap intervals with 10,000 resamples. Holm correction is applied across the four confirmatory contrasts within each model.

| Model           | Bias      | Contrast                    | Mean diff. | 95% CI           | Raw $p$ | Holm $p$ |
|-----------------|-----------|-----------------------------|------------|------------------|---------|----------|
| DeepSeek V4-Pro | length    | Summarization – Append-Only | 0.239      | [−0.085, 0.585]  | 0.221   | 0.662    |
| DeepSeek V4-Pro | length    | RAG+Filter – Append-Only    | 0.150      | [−0.018, 0.322]  | 0.135   | 0.541    |
| DeepSeek V4-Pro | authority | Summarization – Append-Only | −0.161     | [−0.586, 0.196]  | 0.471   | 0.941    |
| DeepSeek V4-Pro | authority | RAG+Filter – Append-Only    | −0.005     | [−0.515, 0.401]  | 0.986   | 0.986    |
| Qwen3.7-Plus    | length    | Summarization – Append-Only | −0.097     | [−0.318, 0.117]  | 0.432   | 1.000    |
| Qwen3.7-Plus    | length    | RAG+Filter – Append-Only    | 0.126      | [−0.126, 0.448]  | 0.436   | 1.000    |
| Qwen3.7-Plus    | authority | Summarization – Append-Only | −0.258     | [−0.435, −0.079] | 0.026   | 0.102    |
| Qwen3.7-Plus    | authority | RAG+Filter – Append-Only    | 0.016      | [−0.420, 0.484]  | 0.951   | 1.000    |

## 6 Discussion

### 6.1 Interpretation of the Round-3 Evidence

The Round-3 evidence changes the interpretation of the architecture comparison. Earlier runs suggested architecture-specific advantages in some cells, but the validator-accepted Round-3 analysis does not provide familywise-confirmed evidence that either Summarization or RAG+Filter improves on Append-Only across the tested model-bias combinations. The strongest directional result, Qwen3.7-Plus authority bias under Summarization, is not a confirmed discovery after Holm correction. We therefore frame the results as evidence against a universal memory-architecture winner and as motivation for deployment-specific measurement.

The cell means still show practically relevant variation. DeepSeek V4-Pro has high authority-bias  $\Gamma_{\text{temporal}}$  under Append-Only and RAG+Filter, while Qwen3.7-Plus shows lower authority-bias  $\Gamma_{\text{temporal}}$  under Summarization. However, these patterns should be read as conditional estimates with uncertainty, not as prescriptive architecture rules. For a production system, the relevant conclusion is not “choose Summarization” or “choose RAG,” but “measure the candidate memory designs on the deployment model, bias axis, and retrieval policy.”

### 6.2 Practical Guidelines

The practitioner-facing recommendation is deliberately conservative. Use Append-Only, Summarization, and RAG+Filter as candidate designs, then run a small paired measurement on the deployment model before selecting a memory architecture. Prefer the design with the lowest measured  $\Gamma_{\text{temporal}}$  only if the uncertainty interval and multiplicity-corrected comparison support that choice. If the intervals overlap or the corrected contrast is not significant, select based on operational criteria such as auditability, latency, cost, and ease of rollback rather than on an unconfirmed bias-resistance claim.

### 6.3 Limitations

- **Recovery provenance.** The final raw logs passed offline validation, but the run required provider-timeout recovery and quarantine/regeneration of empty primary responses. The final logs therefore contain mixed configuration and runner hashes across staged versions.
- **Single contamination rate.** Round-3 focuses on  $p = 0.8$  for the confirmatory rerun. It does not re-estimate the full low-to-high contamination dose-response curve.

- **Model diversity.** The analysis covers DeepSeek V4-Pro and Qwen3.7-Plus. Additional model families would be required before making stronger generalization claims.
- **Synthetic bias injection.** Length expansion and fabricated authority markers are controlled stress tests. They are useful for paired architecture comparison but do not exhaust real deployment bias sources.
- **Fixed architecture hyperparameters.** The RAG threshold  $\theta$  and summarization frequency  $k$  were fixed. Adaptive retrieval or summarization policies may change the architecture ranking.

These limitations narrow the claim rather than invalidate the result. The main Round-3 conclusion is a negative and conditional one: under the preregistered analysis, no architecture contrast survives within-model Holm correction, and any deployment recommendation should be based on local measurement rather than extrapolation from a single table.

### 6.3.1 Threats to Validity

We separate protocol-level limitations (what we did not measure) from interpretation-level threats (how the present measurements could be misread). The former are listed in §?? Limitations; the latter follow. Four threats to validity should be weighed before generalising the present single-round confirmatory analysis.

**Internal validity.** The same scoring pipeline is applied uniformly across all cells; architecture and bias labels are not exposed to the analyst during metric computation. Per-seed  $\Gamma_{\text{temporal}}$  is computed by `gamma_temporal(clean_rows, biased_rows)` without label inspection, and the four-contrast confirmatory family was committed in the protocol before per-seed unblinding. The Round-3 offline validator accepted all 120 seed-condition files and 2,400 solver-completion rows. A reviewer who rejects the four-contrast family can re-aggregate the data into a smaller or larger family; the per-seed values are reported in the Round-3 confirmatory CSV so this re-analysis is possible without re-running the experiment.

**External validity.** The deployment models are DeepSeek V4-Pro and Qwen3.7-Plus, accessed through Alibaba Cloud DashScope and the model providers’ native endpoints, respectively. We do not evaluate closed-weight frontier models (GPT-4o, Claude 4.x, Gemini 2.x) or open-weight models with substantially different post-training regimes (e.g. Llama-3.1-70B-Instruct or a later open-weight release). Whether the negative result—no architecture contrast survives within-model Holm correction—holds for those families is an empirical question our design cannot answer.

**Construct validity.**  $\Gamma_{\text{temporal}}$  is the Wasserstein-1 distance between biased and clean output-length distributions after z-scoring by the clean arm. This metric captures length-distribution shift but not, for example, content-fidelity shift (covered by  $\Gamma_{\text{content}}$  for the summarization architecture) or retrieval-shaping (covered by  $\Gamma_{\text{retrieval}}$ ). A reader who treats low  $\Gamma_{\text{temporal}}$  as a complete bias-resistance guarantee would over-interpret the present results;  $\Gamma_{\text{temporal}}$  is a necessary but not sufficient proxy for bias-resistance.

**Conclusion validity.** The within-model Holm correction at  $\alpha = .05$  is the preregistered threshold. The single contrast that approaches this threshold under Qwen3.7-Plus authority bias (mean paired difference =  $-0.258$ , raw  $p = 0.0255$ , Holm-adjusted  $p = 0.102$ ) is reported as a directional observation, not as evidence of an effect. A reviewer who lowers  $\alpha$  to  $.10$  would treat this contrast as a candidate finding; we do not, because the family-level decision rule was fixed before the per-seed unblinding.

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## 7 Broader Impact

Memory architecture choices affect what information future agents can see, compress, and retrieve. A false claim that one architecture is universally safer could encourage overconfident deployment. The Round-3 result argues for the opposite norm: memory systems should be instrumented, compared under paired bias-injection tests, and monitored as model providers or prompts change. This is especially important because the recovery history shows that provider behavior and empty responses can affect live evidence collection unless the pipeline validates raw outputs before analysis.

## 8 Conclusion

We compared three memory architectures for bias resistance across DeepSeek V4-Pro and Qwen3.7-Plus under length and authority bias at  $p = 0.8$ . The final Round-3 collection contains 120 seed-condition JSONL files and 2,400 solver-completion rows and passed the project validator. No confirmatory contrast survived within-model Holm correction. The strongest directional contrast was Qwen3.7-Plus authority bias under Summarization (mean difference =  $-0.258$ , raw  $p = 0.0255$ , Holm  $p = 0.102$ ), which is suggestive but not confirmatory. We therefore conclude that memory-architecture effects are model- and bias-conditional, and that practitioners should validate candidate architectures on their own deployment setting rather than assuming a universal winner.

### Future Work

The Round-3 design intentionally narrows scope to enable clean within-model Holm correction; several extensions would close the gap between the present single-confirmatory-round result and a deployment-grade memory-architecture recommendation.

(i) **Full dose-response curve at  $p \in \{0.2, 0.5, 0.8\}$ .** The canonical registry records that DeepSeek V4-Pro length bias was originally planned at three contamination rates; Round-3 confirmatory analysis was restricted to  $p = 0.8$  to limit the multiple-comparison burden. As a partial prior, the 2026-07-08 R2 pilot collected dose-response data on DeepSeek V4-Chat at  $p=0.2$  and  $p=0.5$  with  $T=30$  rounds ( $n=10$  seeds, 60 cells; `memory_architecture/experiments/results/dose_response.json`); those cells are not directly comparable to the present Round-3 V4-Pro  $p=0.8$  data because the model and round count differ, but they document that the architecture ranking at lower contamination pressures does not invert. Re-running the lower-rate cells at  $n=10$  paired seeds under the same V4-Pro  $T=10$  protocol would close the gap and enable a model- and rate-conditional picture of when architecture choice matters.

(ii) **Broader model families and on-policy fine-tuning.** The present analysis is restricted to DeepSeek V4-Pro and Qwen3.7-Plus under the full Round-3 protocol. A small *schema-portability pilot* on `qwen3-max` (DashScope) at  $p=0.8$ , length bias, Append-Only,  $n=10$  seeds confirms the data-acquisition code is portable to other Qwen-family deployments without protocol change (`paper5_qwen3max_pilot_20260714/`); the pilot uses a simplified single-step inference (no memory retrieval, no summariser) so its  $\Gamma_{\text{temporal}}$  values are not directly comparable to the Round-3 numbers in Table 3 and are reported only to document that the JSONL schema and the offline `gamma_temporal` function accept new model endpoints without code change. Extending to a third open-weight family (e.g. Llama-3.1-70B-Instruct or a later open-weight release) and to a closed-weight frontier model (e.g. Claude 4 Sonnet, GPT-4o) under the *full* Round-3 protocol would test (a) whether the negative result generalises across post-training regimes, and (b) whether the TTRL boundary statement

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in §?? holds under actual on-policy training. The Bay & Yearick (2026) ceiling becomes directly relevant for closed-weight APIs whose selection accuracy under repeated sampling is unobservable.

**(iii) Adaptive architecture hyperparameters.** Both  $\theta$  (RAG threshold) and  $k$  (summarisation frequency) were held fixed across the confirmatory cells. A follow-on sweep that treats the architecture hyperparameters as first-class inputs—and reports the per-cell architecture ranking across the  $\theta \times k$  grid—would convert the present conditional conclusion into a deployment-specific recommendation.

The Round-3 analysis is regenerated from validator-accepted raw JSONL without additional network calls. The final collection contains 120 seed-condition files and 2,400 solver-completion rows. The validator report records `accepted: true`; the manifest-level audit reports no partial files, no missing required fields, no empty primary responses, and no blank request identifiers.

**Implementation environment.** The offline validator and analysis scripts use Python 3.10+, NumPy 1.24+, SciPy 1.10+, scikit-learn 1.3+, and transformers 4.x for tokenizer-side utilities. No model weights are loaded locally.

**Round-3 artifacts.** The dated evidence package includes the final validator report, final collection validation manifest, confirmatory results CSV, confirmatory analysis manifest, empty-response quarantine manifest, and recovery/deviation record. These artifacts document the provider-timeout recovery, empty-response quarantine and regeneration, mixed provenance hashes, and offline recomputation of  $\Gamma_{\text{temporal}}$ .

**Random seed handling.** Round-3 uses ten paired seeds per cell. Seeds are matched across architectures within the same model and bias type so that architecture contrasts are paired at the seed level. Provider request identifiers and retry metadata are retained in the raw logs to support auditability.

**Learning rate.** PAPER5 does not fine-tune or train any model. All reported results are obtained through inference-only solver and memory-update calls, so no learning rate, optimizer, or gradient-based update is applicable.

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## A Round-3 Evidence Package

The confirmatory evidence package is external to the TeX source and should be archived with the manuscript. The key files are:

- `paper5_round3_validator_report_final_20260714.json`
- `paper5_round3_final_collection_validation_20260714.json`
- `paper5_round3_confirmatory_results_20260714.csv`
- `paper5_round3_confirmatory_analysis_manifest_20260714.json`
- `paper5_round3_empty_response_quarantine_manifest_20260714.json`
- `paper5_round3_deviation_record_20260714.md`

The manuscript results in Tables 3 and 4 are the manuscript rendering of that evidence package.

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